POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6043B | Semester VI | Theory | 4 Credits

Microcontrollers & Applications

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6043B as Microcontrollers & Applications in Semester VI for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kanpur’s Microcontrollers and Applications course and polytechnic microcontroller curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Firmware designs, circuit schematics, timing analysis and embedded systems must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

> **Verified source note:** Verified sources support 8051 architecture, programming, interfacing, and advanced microcontroller concepts. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Microcontrollers & Applications studies the architecture, programming and interfacing of small-scale computing systems in electronic devices. It develops the ability to analyze 8051 and advanced (AVR/ARM) microcontroller architectures, evaluate Embedded C programming for peripheral control, understand interrupt and timing systems, and coordinate embedded solutions while respecting the technical and timing boundaries of electronic and computer engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Digital electronics, microprocessors (8085/8086), basic C programming and electronic circuits.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the architecture, memory organization and registers of the 8051 microcontroller.
2. Apply assembly and Embedded C programming to implement logic and control functions.
3. Analyze the operation and programming of timers, interrupts and serial communication modules.
4. Explain the interfacing of microcontrollers with external peripherals and understand advanced (AVR/ARM) architectures.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the architecture and memory organization of the 8051 microcontroller.	Understand
CO2	Develop programs in assembly and Embedded C for microcontroller-based tasks.	Apply
CO3	Analyze and program timers, interrupts and serial communication in microcontrollers.	Apply
CO4	Interfacing microcontrollers with external peripherals and understand advanced architectures.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	8051 Microcontroller Architecture	Comparison between Microprocessor and Microcontroller; 8051 block diagram; CPU and Registers; Memory organization: Program and Data memory; I/O ports; Special Function Registers (SFRs).	Approx. 14 hours	CO1	Module 1
2	8051 Programming and Instructions	Addressing modes: Immediate, Register, Direct, Indirect; Instruction set: Data transfer, Arithmetic, Logical, Branching; Introduction to Embedded C; Writing and debugging programs.	Approx. 16 hours	CO2	Module 2
3	Timers, Interrupts and Serial Communication	Programming 8051 Timers/Counters: TMOD, TCON registers; Timer modes 0, 1, 2; Interrupt structure: IE, IP registers; Serial communication: SBUF, SCON registers, Baud rate calculation.	Approx. 16 hours	CO3	Module 3
4	Interfacing and Advanced Architectures	Interfacing LEDs, LCD, Keypad; Interfacing Sensors (Temperature, Ultrasonic); Motor control: Stepper and Servo; Introduction to AVR (ATmega328) and ARM Cortex-M; IoT basics.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

8051 Microcontroller Architecture

Comparison between Microprocessor and Microcontroller; 8051 block diagram; CPU and Registers; Memory organization: Program and Data memory; I/O ports; Special Function Registers (SFRs).

CO1 · Approx. 14 hours

8051 Programming and Instructions

Addressing modes: Immediate, Register, Direct, Indirect; Instruction set: Data transfer, Arithmetic, Logical, Branching; Introduction to Embedded C; Writing and debugging programs.

CO2 · Approx. 16 hours

Timers, Interrupts and Serial Communication

Programming 8051 Timers/Counters: TMOD, TCON registers; Timer modes 0, 1, 2; Interrupt structure: IE, IP registers; Serial communication: SBUF, SCON registers, Baud rate calculation.

CO3 · Approx. 16 hours

Interfacing and Advanced Architectures

Interfacing LEDs, LCD, Keypad; Interfacing Sensors (Temperature, Ultrasonic); Motor control: Stepper and Servo; Introduction to AVR (ATmega328) and ARM Cortex-M; IoT basics.

CO4 · Approx. 14 hours

MODULE 1

8051 Microcontroller Architecture

Comparison between Microprocessor and Microcontroller; 8051 block diagram; CPU and Registers; Memory organization: Program and Data memory; I/O ports; Special Function Registers (SFRs).

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

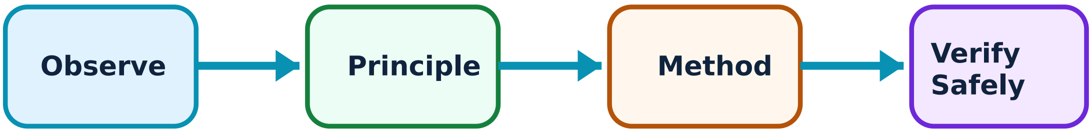
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for 8051 Microcontroller Architecture: identify the system, state its principle, follow the correct method and verify the result safely.

1. Introduction and CPU Architecture−

Microcontrollers integrate CPU, memory, and I/O on a single chip. The 8051 is a classic 8-bit architecture with an accumulator-based CPU. It features specific registers like B, DPTR, and PC, along with a Program Status Word (PSW) containing flags.

Study and application method

Differentiate between 'Von Neumann' and 'Harvard' architectures; list four internal components of the '8051' block diagram.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Von Neumann' and 'Harvard' architectures; list four internal components of the '8051' block diagram.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Improper register usage can lead to data corruption. Do not use SFRs without authorised verification of their specific functions and bit-level addresses.

2. Memory and I/O Organization−

The 8051 has separate address spaces for program and data memory. Internal RAM includes bit-addressable and byte-addressable areas. Four 8-bit I/O ports (P0-P3) are used for interfacing, with some ports having dual functions like address/data multiplexing.

Study and application method

Explain the 'Internal RAM' map of 8051 including the bit-addressable area; describe the dual functions of 'Port 3' pins.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'Internal RAM' map of 8051 including the bit-addressable area; describe the dual functions of 'Port 3' pins.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Excessive current draw from I/O ports can damage the chip. Do not interface high-current loads without authorised drivers, buffers, or opto-isolators.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

8051 Programming and Instructions

Addressing modes: Immediate, Register, Direct, Indirect; Instruction set: Data transfer, Arithmetic, Logical, Branching; Introduction to Embedded C; Writing and debugging programs.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

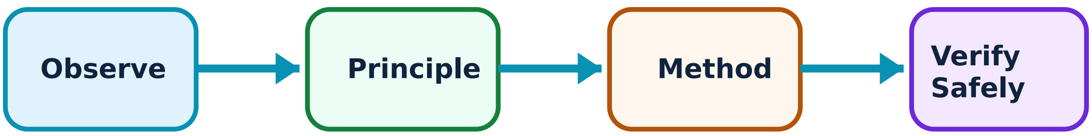
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for 8051 Programming and Instructions: identify the system, state its principle, follow the correct method and verify the result safely.

1. Assembly Language Programming−

8051 assembly uses mnemonics to represent machine instructions. Addressing modes define how the CPU accesses data. Logic instructions (AND, OR, XOR) and branching (LJMP, SJMP, ACALL) are essential for program flow control.

Study and application method

Write a Verilog-like pseudo-code or assembly snippet to 'Add two 8-bit numbers' in 8051; explain the 'Indirect Addressing' mode.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write a Verilog-like pseudo-code or assembly snippet to 'Add two 8-bit numbers' in 8051; explain the 'Indirect Addressing' mode.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Incorrect branching can lead to stack overflow or infinite loops. All assembly programs must follow authorised stack management and subroutine return protocols.

2. Embedded C Programming−

Embedded C is preferred for large projects due to its portability and ease of maintenance. It uses specific keywords (like sbit, sfr) to access 8051 hardware. Compilers like Keil convert C code into HEX files for burning onto the microcontroller.

Study and application method

Compare 'Assembly' and 'Embedded C' for microcontroller programming; write a simple C program to 'Toggle a pin' at a specific port.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Assembly' and 'Embedded C' for microcontroller programming; write a simple C program to 'Toggle a pin' at a specific port.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Standard C libraries may be too large for small microcontrollers. Do not use resource-intensive functions (like printf) without authorised memory analysis and optimization.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Timers, Interrupts and Serial Communication

Programming 8051 Timers/Counters: TMOD, TCON registers; Timer modes 0, 1, 2; Interrupt structure: IE, IP registers; Serial communication: SBUF, SCON registers, Baud rate calculation.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

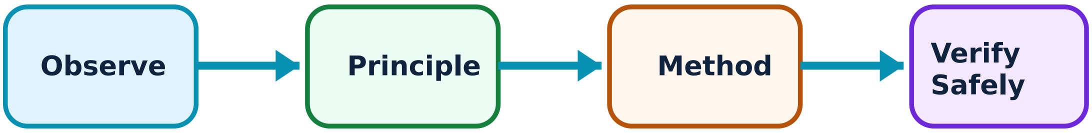
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Timers, Interrupts and Serial Communication: identify the system, state its principle, follow the correct method and verify the result safely.

1. Timers and Counters−

8051 has two 16-bit timers (T0 and T1) that can operate as timers (internal clock) or counters (external pulses). TMOD defines the mode (e.g., Mode 2 for auto-reload), and TCON controls the start/stop and flags.

Study and application method

Explain the function of the 'TMOD' register; calculate the count value for a 1ms delay conceptually using a 12MHz crystal.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the function of the 'TMOD' register; calculate the count value for a 1ms delay conceptually using a 12MHz crystal.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Timer overflow can cause timing inaccuracies if not handled promptly. Do not use software-based delays for precise timing without authorised hardware timer synchronization.

2. Interrupts and UART−

Interrupts allow the microcontroller to respond to urgent events. 8051 has five interrupts: two external, two timers, and one serial. Serial communication (UART) uses SBUF for data and SCON for control, with Timer 1 usually setting the baud rate.

Study and application method

List the five 'Interrupts' of 8051 in order of their default priority; describe the role of the 'SBUF' register in serial communication.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the five 'Interrupts' of 8051 in order of their default priority; describe the role of the 'SBUF' register in serial communication.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Uncontrolled interrupts can crash the system. All interrupt service routines (ISRs) must follow authorised priority settings and be kept as short as possible.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Interfacing and Advanced Architectures

Interfacing LEDs, LCD, Keypad; Interfacing Sensors (Temperature, Ultrasonic); Motor control: Stepper and Servo; Introduction to AVR (ATmega328) and ARM Cortex-M; IoT basics.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

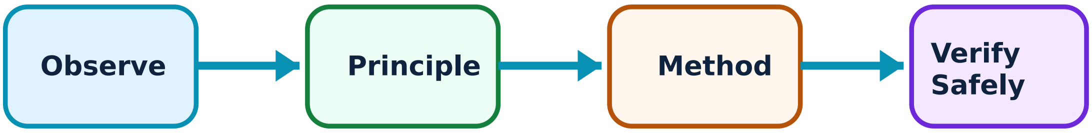
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Interfacing and Advanced Architectures: identify the system, state its principle, follow the correct method and verify the result safely.

1. Peripheral Interfacing−

Interfacing involves connecting the microcontroller to external world devices. LCDs require data and control lines (RS, EN). Keypads use a matrix scanning technique. Motors require H-bridge or specialized drivers to handle power.

Study and application method

Sketch the interface of a '16x2 LCD' with 8051; explain the 'Matrix Keypad' scanning principle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the interface of a '16x2 LCD' with 8051; explain the 'Matrix Keypad' scanning principle.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□□.

Common mistake / precaution: Floating inputs can cause erratic behavior. Do not interface sensors or switches without authorised pull-up/pull-down resistors and debouncing circuits.

2. Advanced Microcontrollers and IoT−

AVR and ARM architectures offer higher performance, more memory, and advanced peripherals (ADC, PWM, SPI, I2C). They are the backbone of modern IoT devices. IoT involves connecting these microcontrollers to the internet for remote monitoring and control.

Study and application method

Compare '8051' and 'ARM Cortex-M' architectures conceptually; describe the role of a 'Wi-Fi Module' (e.g., ESP8266) in an IoT system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare '8051' and 'ARM Cortex-M' architectures conceptually; describe the role of a 'Wi-Fi Module' (e.g., ESP8266) in an IoT system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Advanced microcontrollers have complex clock and power configurations. All advanced designs must follow authorised hardware abstraction layers (HAL) and power-up sequences.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: 8051 Microcontroller Architecture

Use the concept flow to revise observation, principle, method and verification.

Module 2: 8051 Programming and Instructions

Use the concept flow to revise observation, principle, method and verification.

Module 3: Timers, Interrupts and Serial Communication

Use the concept flow to revise observation, principle, method and verification.

Module 4: Interfacing and Advanced Architectures

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Write an Embedded C program to blink an LED connected to Port 1 with a 500ms delay.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a Verilog-like logic for a 'BCD to Seven Segment' decoder for 8051 interfacing.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the timing diagram for 8051 serial communication showing Start, Data, and Stop bits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the TMOD value for Timer 0 in Mode 1 (16-bit timer) conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a microcontroller in a household appliance (e.g., Microwave, Washing machine) and list its likely functions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — 8051 Microcontroller Architecture

Explain Introduction and CPU Architecture and state one correct method, application or precaution.−

Microcontrollers integrate CPU, memory, and I/O on a single chip. The 8051 is a classic 8-bit architecture with an accumulator-based CPU. It features specific registers like B, DPTR, and PC, along with a Program Status Word (PSW) containing flags.

Answer framework: Differentiate between 'Von Neumann' and 'Harvard' architectures; list four internal components of the '8051' block diagram.

Important point: Improper register usage can lead to data corruption. Do not use SFRs without authorised verification of their specific functions and bit-level addresses.

Explain Memory and I/O Organization and state one correct method, application or precaution.−

The 8051 has separate address spaces for program and data memory. Internal RAM includes bit-addressable and byte-addressable areas. Four 8-bit I/O ports (P0-P3) are used for interfacing, with some ports having dual functions like address/data multiplexing.

Answer framework: Explain the 'Internal RAM' map of 8051 including the bit-addressable area; describe the dual functions of 'Port 3' pins.

Important point: Excessive current draw from I/O ports can damage the chip. Do not interface high-current loads without authorised drivers, buffers, or opto-isolators.

Module 2 — 8051 Programming and Instructions

Explain Assembly Language Programming and state one correct method, application or precaution.−

8051 assembly uses mnemonics to represent machine instructions. Addressing modes define how the CPU accesses data. Logic instructions (AND, OR, XOR) and branching (LJMP, SJMP, ACALL) are essential for program flow control.

Answer framework: Write a Verilog-like pseudo-code or assembly snippet to 'Add two 8-bit numbers' in 8051; explain the 'Indirect Addressing' mode.

Important point: Incorrect branching can lead to stack overflow or infinite loops. All assembly programs must follow authorised stack management and subroutine return protocols.

Explain Embedded C Programming and state one correct method, application or precaution.−

Embedded C is preferred for large projects due to its portability and ease of maintenance. It uses specific keywords (like sbit, sfr) to access 8051 hardware. Compilers like Keil convert C code into HEX files for burning onto the microcontroller.

Answer framework: Compare 'Assembly' and 'Embedded C' for microcontroller programming; write a simple C program to 'Toggle a pin' at a specific port.

Important point: Standard C libraries may be too large for small microcontrollers. Do not use resource-intensive functions (like printf) without authorised memory analysis and optimization.

Module 3 — Timers, Interrupts and Serial Communication

Explain Timers and Counters and state one correct method, application or precaution. −

8051 has two 16-bit timers (T0 and T1) that can operate as timers (internal clock) or counters (external pulses). TMOD defines the mode (e.g., Mode 2 for auto-reload), and TCON controls the start/stop and flags.

Answer framework: Explain the function of the 'TMOD' register; calculate the count value for a 1ms delay conceptually using a 12MHz crystal.

Important point: Timer overflow can cause timing inaccuracies if not handled promptly. Do not use software-based delays for precise timing without authorised hardware timer synchronization.

Explain Interrupts and UART and state one correct method, application or precaution. −

Interrupts allow the microcontroller to respond to urgent events. 8051 has five interrupts: two external, two timers, and one serial. Serial communication (UART) uses SBUF for data and SCON for control, with Timer 1 usually setting the baud rate.

Answer framework: List the five 'Interrupts' of 8051 in order of their default priority; describe the role of the 'SBUF' register in serial communication.

Important point: Uncontrolled interrupts can crash the system. All interrupt service routines (ISRs) must follow authorised priority settings and be kept as short as possible.

Module 4 — Interfacing and Advanced Architectures

Explain Peripheral Interfacing and state one correct method, application or precaution.−

Interfacing involves connecting the microcontroller to external world devices. LCDs require data and control lines (RS, EN). Keypads use a matrix scanning technique. Motors require H-bridge or specialized drivers to handle power.

Answer framework: Sketch the interface of a '16x2 LCD' with 8051; explain the 'Matrix Keypad' scanning principle.

Important point: Floating inputs can cause erratic behavior. Do not interface sensors or switches without authorised pull-up/pull-down resistors and debouncing circuits.

Explain Advanced Microcontrollers and IoT and state one correct method, application or precaution.−

AVR and ARM architectures offer higher performance, more memory, and advanced peripherals (ADC, PWM, SPI, I2C). They are the backbone of modern IoT devices. IoT involves connecting these microcontrollers to the internet for remote monitoring and control.

Answer framework: Compare '8051' and 'ARM Cortex-M' architectures conceptually; describe the role of a 'Wi-Fi Module' (e.g., ESP8266) in an IoT system.

Important point: Advanced microcontrollers have complex clock and power configurations. All advanced designs must follow authorised hardware abstraction layers (HAL) and power-up sequences.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: 8051 Microcontroller Architecture.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: 8051 Programming and Instructions.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Timers, Interrupts and Serial Communication.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Interfacing and Advanced Architectures.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Comparison between Microprocessor and Microcontroller; 8051 block diagram; CPU and Registers; Memory organization: Program and Data memory; I/O ports; Special Function Registers (SFRs)..
2. Develop a complete answer or practical plan for: Addressing modes: Immediate, Register, Direct, Indirect; Instruction set: Data transfer, Arithmetic, Logical, Branching; Introduction to Embedded C; Writing and debugging programs..
3. Develop a complete answer or practical plan for: Programming 8051 Timers/Counters: TMOD, TCON registers; Timer modes 0, 1, 2; Interrupt structure: IE, IP registers; Serial communication: SBUF, SCON registers, Baud rate calculation..
4. Develop a complete answer or practical plan for: Interfacing LEDs, LCD, Keypad; Interfacing Sensors (Temperature, Ultrasonic); Motor control: Stepper and Servo; Introduction to AVR (ATmega328) and ARM Cortex-M; IoT basics..

LAST-MILE REVISION

Quick Revision

Module 1: 8051 Microcontroller Architecture

Comparison between Microprocessor and Microcontroller; 8051 block diagram; CPU and Registers; Memory organization: Program and Data memory; I/O ports; Special Function Registers (SFRs).

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: 8051 Programming and Instructions

Addressing modes: Immediate, Register, Direct, Indirect; Instruction set: Data transfer, Arithmetic, Logical, Branching; Introduction to Embedded C; Writing and debugging programs.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Timers, Interrupts and Serial Communication

Programming 8051 Timers/Counters: TMOD, TCON registers; Timer modes 0, 1, 2; Interrupt structure: IE, IP registers; Serial communication: SBUF, SCON registers, Baud rate calculation.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Interfacing and Advanced Architectures

Interfacing LEDs, LCD, Keypad; Interfacing Sensors (Temperature, Ultrasonic); Motor control: Stepper and Servo; Introduction to AVR (ATmega328) and ARM Cortex-M; IoT basics.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Introduction and CPU Architecture	Microcontrollers integrate CPU, memory, and I/O on a single chip.
Memory and I/O Organization	The 8051 has separate address spaces for program and data memory.
Assembly Language Programming	8051 assembly uses mnemonics to represent machine instructions.
Embedded C Programming	Embedded C is preferred for large projects due to its portability and ease of maintenance.
Timers and Counters	8051 has two 16-bit timers (T0 and T1) that can operate as timers (internal clock) or counters (external pulses).
Interrupts and UART	Interrupts allow the microcontroller to respond to urgent events.
Peripheral Interfacing	Interfacing involves connecting the microcontroller to external world devices.
Advanced Microcontrollers and IoT	AVR and ARM architectures offer higher performance, more memory, and advanced peripherals (ADC, PWM, SPI, I2C).

LEARNING RESOURCES

References

Recommended microcontroller references

1. Muhammad Ali Mazidi, Janice Gillispie Mazidi and Rolin D. McKinlay, The 8051 Microcontroller and Embedded Systems.
2. Kenneth J. Ayala, The 8051 Microcontroller.
3. Ajay V. Deshmukh, Microcontrollers: Theory and Applications.
4. Raj Kamal, Embedded Systems: Architecture, Programming and Design.

Approved public microcontroller sources

- https://nptel.ac.in/courses/117104072
- https://nptel.ac.in/courses/108105102

These sources provide the verified study basis while official Course 6043B detail is unavailable; they do not replace official firmware designs, authorised safety-critical systems, certified equipment specifications or authorised electronic product plans.